

TESTs - Lesson 20 - Lottery

Which command is used to install Hardhat as a local development dependency?

Environment setup for VS Code

- For our tests extension to work, we need to install the Hardhat extension.
- Create a file named `hardhat.config.js` in the root of the project.

• Run `npm install --save-dev hardhat` to install Hardhat as a development dependency.

- A `npm init hardhat`
- B `npm install hardhat -g`
- C `npm install --save-dev hardhat`

Who is responsible for defining the initial betting price and fee for the lottery?

Implementation details

- Implement `ownable`.
- Owner deploy lottery.
- Owner start lottery.
- Define a block range.
- Players must buy an ERC20 token.
- Players pay ERC20 to bet.

- A The first player who places a bet.
- B The contract owner upon deployment.
- C They are hardcoded values that cannot be changed.

What must a player do before they can place a bet in the lottery?

Articles in this topic

Understanding my Method Call Breakdown (in "method" tab)

Understanding the 10,000 Block Range Used for drawing Lots

- A Send ETH directly to the lottery contract.
- B Obtain the specific ERC20 token used for betting.
- C Register their wallet address with the contract owner.

What is the distinction between the 'price' and the 'fee' associated with each bet?

Understanding my Method Call Breakdown (in "method" tab)

Understanding the 10,000 Block Range Used for drawing Lots

Understanding my Method Call Breakdown (in "method" tab)

- A The 'price' goes to the owner, and the 'fee' goes to the prize pool.
- B The 'price' goes to the prize pool, and the 'fee' goes to the owner's pool.
- C Both the 'price' and the 'fee' are added to the main prize pool for the winner.

What is the primary architectural difference between the `Lottery.sol` and `LotteryToken.sol` contracts?

- A `Lottery.sol` handles the game logic, while `LotteryToken.sol` is the ERC20 token for payments.
- B `Lottery.sol` is the main contract, and `LotteryToken.sol` is a library it imports for math functions.
- C They are two independent contracts that do not interact with each other.

According to the implementation details, how is the `LotteryToken` contract created?



- A It is deployed separately, and its address is passed to the Lottery contract's constructor.
- B The `Lottery` contract deploys the `LotteryToken` contract from within its own constructor.

What action must the owner take to make the lottery operational after deployment?



- A Fund the contract with an initial prize amount.
- B Call a function to start the lottery and define a closing timestamp.
- C Manually approve each player's wallet.

Why is the function to determine the winner (roll the lottery) designed to be callable by anyone?



- A To allow any player to check if they have won at any time.
- B To prevent a denial of service attack where the owner could stop the lottery from concluding.
- C To distribute the gas cost of rolling the lottery among all participants.

How is the winner of the lottery determined in this contract design?



- A The player who bets the largest amount wins.
- B The contract owner manually selects the winner.
- C A random number is generated using RANDAO to select a winner from the list of bets.